Computer Networks
Fall 2024/25
SOLUTION
Exercise-3-solution

Problem 1 (20 points).
1. The modified protocol is correct but is less efficient.
2. 1. 16 messages. The messages are from 135 until 150.
2. 30 messages. From 121 until 150.

Problem 2 (20 points).
Answer: 3.558 days
The size of the sequence space is 2^58 bytes. A 7.5 Tbps transmitter uses up sequence space at a rate of .9375 x 10^12 sequence numbers per second. It takes around **307,446 seconds** to wrap around. Since there are 86,400 seconds in a day, it takes **3.558 days** to wrap around (even at 7.5 Tbps).

Problem 3 (20 points).

#	Without buffering	With buffering
1	6301	6601
2	6101	6601
3	6101	6601
4	6301	7001
5	6301	6301

Problem 4 (20 points).

question	Answer (in msec)
1	**15**
2	**27**
3	**17**
4	**20**
5	**23**
6	**40**

6. Ack for the first packet arrives at 8.
The timer restarts at 8 and expires at 18.
Resubmit the second packet at 18. The second packet arrives (half RTT) at 22.
Ack for the second packet sent at 22, arrive at 26.
The timer restarts at 26 and expires at 36.
Resubmit the third packet at 36.
The third packet arrives (half RTT) at **40**.

Problem 5 (20 points).

a. TCP slowstart is operating in the intervals [1,4] , [13,16], [25,28]and [34,36]. Congestion avoidance is operating in the intervals [4,12] , [16,24] and [29,33].

b. Only after the 28th transmission round, packet loss is recognized by a triple duplicate.

c. After the12th, 24th and 33nd segment loss is detected due to timeout (hence congestion window size is 1).

d. The threshold is initially 8 (then again 8 twice) then 4 (then 4 again).

e. Packet 75 is sent in the 10th transmission round.